

SecureFinger AI

Contactless Fingerprint Anti-Spoofing System

Technical Evaluation Report

Answers to 7 Evaluation Questions

Model: MobileNetV3-Small | Epochs: 25 | Best Val Loss: 0.0

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Q1: Why did you choose MobileNetV3-Small as the backbone?

MobileNetV3-Small was selected for the following reasons:

1. Efficiency: It is designed for mobile and edge deployment with a small parameter count (approximately 2.5M parameters), making it suitable for real-time biometric systems.
2. Performance: Despite its small size, MobileNetV3-Small achieves competitive accuracy on image classification benchmarks due to its carefully designed architecture combining NAS-derived search and squeeze-and-excitation attention.
3. Transfer Learning Compatibility: Pretrained on ImageNet, the early convolutional layers already learn generic visual features (edges, textures, patterns) that transfer well to fingerprint images.
4. Speed: Inference is fast enough for real-time PAD applications, processing images in milliseconds on CPU, which is critical for production deployment.

Q2: How does the data augmentation strategy help prevent overfitting?

The augmentation pipeline applies four transformations during training:

- Rotation (up to 15 degrees): Simulates variations in fingerprint orientation during capture.
- Brightness variation (20% range): Mimics different lighting conditions in real-world deployment.
- Horizontal flip (50% probability): Doubles the effective dataset size and adds invariance to left-right orientation.
- Gaussian noise (std=0.05): Simulates sensor noise and compression artifacts, making the model robust to input perturbations.

These augmentations increase the effective diversity of the training set, preventing the model from memorizing specific visual artifacts of individual training images and forcing it to learn general spoof-detection features.

Q3: Explain the APCER, BPCER, and EER metrics and why BPCER=3% is the target.

APCER (Attack Presentation Classification Error Rate) measures the fraction of SPOOF samples incorrectly classified as LIVE. A low APCER means the system rarely misses attacks.

BPCER (Bona Fide Presentation Classification Error Rate) measures the fraction of LIVE samples incorrectly classified as SPOOF. A low BPCER means genuine users are rarely rejected.

EER (Equal Error Rate) is the point where APCER equals BPCER. It provides a single-number summary of the model's discriminative ability.

BPCER=3% is chosen as the target because in security-critical biometric systems, the cost of accepting a spoof (false negative) is much higher than rejecting a genuine user (false positive). A 3% BPCER ensures that at most 3 out of 100 genuine users are incorrectly rejected, while minimizing the risk of spoof acceptance. This threshold represents a practical security/UX trade-off.

Q4: How does transfer learning improve performance with limited data?

Transfer learning leverages knowledge learned from a large dataset (ImageNet, 1.2M images) and transfers it to the target task (fingerprint PAD). The pretrained backbone already understands:

- Low-level features: edges, corners, textures (from early convolutional layers)
- Mid-level features: patterns, shapes, and structural elements (from middle layers)
- High-level semantic features: object parts and compositions (from deeper layers)

With only 120 training images, training from scratch would lead to severe overfitting. By freezing the pretrained feature extractor and only training the classification head, we effectively have millions of learned features guiding the model,

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dramatically reducing the number of trainable parameters and preventing overfitting.

Q5: Describe the Grad-CAM explainability approach and its value.

Grad-CAM (Gradient-weighted Class Activation Mapping) visualizes which regions of the input image most influenced the model's prediction. It works by:

1. Performing a forward pass to get the predicted class score.
2. Computing gradients of the target class score with respect to the final convolutional layer feature maps.
3. Global-average-pooling these gradients to obtain importance weights for each feature map.
4. Weighting the feature maps by these weights and summing them to produce a heatmap.
5. Applying ReLU to retain only positive contributions.

Value: Grad-CAM provides transparency into the model's decision-making process. For fingerprint PAD, it shows whether the model is attending to genuine fingerprint ridge patterns (LIVE) or artifacts like printing marks, edges, or smooth regions (SPOOF). This is critical for security auditing and building trust in the system.

Q6: How would you deploy this model in a production fingerprint authentication pipeline?

In production, the PAD module would serve as a security gate (FP-02) in the authentication pipeline:

1. Capture: The fingerprint sensor captures a fingerprint image.
2. Preprocessing: The image is resized to 224x224, normalized with ImageNet statistics.
3. Inference: The model outputs a probability score for LIVE vs SPOOF.
4. Threshold Decision: If the SPOOF probability exceeds the calibrated threshold (BPCER=3%), the authentication is rejected.
5. Logging: All decisions, scores, and Grad-CAM visualizations are logged for audit trails.

Deployment considerations:

- Model serving via TorchScript or ONNX for optimized inference.
- GPU acceleration for real-time throughput.
- Continuous monitoring of APCER/BPCER in production to detect model drift.
- Periodic retraining with new spoof samples as attack techniques evolve.

Q7: What are the limitations of this prototype and how would you improve it?

Limitations:

1. Synthetic data: The current dataset uses procedurally generated fingerprint patterns, which may not capture the full diversity of real-world fingerprints and spoof materials.
2. Small dataset: 120 images is far fewer than typical PAD datasets (thousands of images).
3. Single model: An ensemble of models or a more complex architecture could improve robustness.
4. Static threshold: The threshold is calibrated once; in production, adaptive thresholds based on environmental conditions would be beneficial.

Improvements:

1. Collect real fingerprint images and spoof samples (silicone, gelatin, printed, etc.).
2. Use data augmentation more aggressively and consider generative augmentation (GANs).
3. Implement test-time augmentation (TTA) for more robust predictions.
4. Add multi-scale feature extraction to handle varying fingerprint resolutions.
5. Implement continuous learning to adapt to new spoof materials over time.
6. Add presentation attack instrumentation (PAI) metadata as additional input features.

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Model Performance Summary

Evaluation Results (from evaluation_results.json):

Threshold at BPCER ~3%: 0.01

APCER at threshold: 0.0

BPCER at threshold: 0.0

EER: 0.0

EER Threshold: 0.9391281008720398

Total Test Samples: 18

Note: The synthetic dataset produces near-perfect classification. Real-world performance will vary based on data quality and diversity.